

CS1005 - DISCRETE STRUCTURES

Quiz No. 2

Name:

Roll No:

Section: A

Total Marks: 10

Q1. For each of the following formulas write a formula f (using quantifiers) expressing the formula, find a formula in negation normal form equivalent to $\sim f$, and express the meaning of the negation in words. **(3 Marks)**

(a) For every x and for every y , $x + y = y + x$

(b) For some y , $2x^2 + 1$ is always greater than x^2y

(c) For every x and y , there is a z where $2z = x + y$

NOTE: Negation Normal Form means negating the formula and pushing the $\sim$ symbol inward until it is only in front of the variables/predicates.

Q2. Let U be the set of all problems on a comprehensive list of problems in science. Define four predicates over U by:

$P(x)$: x is a mathematics problem

$Q(x)$: x is difficult

$R(x)$: x is easy

$S(x)$: x is unsolvable

Translate into English sentences each of the following formulas: **(2 Marks)**

(a) $\forall x [P(x) \rightarrow (R(x) \vee S(x))]$

(b) $\forall x [P(x) \rightarrow (Q(x) \leftrightarrow \neg R(x))]$

Q3. Let $P(x)$, $Q(x)$, and $R(x)$ be the statements “ x is a clear explanation,” “ x is satisfactory,” and “ x is an excuse,” respectively. Suppose that the domain for x consists of all English text. Express statements a , b and c using quantifiers, logical connectives and $P(x)$, $Q(x)$, and $R(x)$.

(5 Marks)

(a) All clear explanations are satisfactory.

(b) Some excuses are unsatisfactory.

(c) Some excuses are not clear explanations.

(d) Does (c) follow from (a) and (b)?